

Table 1: **Language generation.** The number of correct answers to 100 synthetic BPs, 100 selected BPs from each of Bongard HOI and Bongard-OpenWorld, and all 60 BPs from Bongard-RWR. Three main strategies: Direct, Descriptive, and Contrastive, denoted as Di, De, and Co, resp., are considered. The best result for a given strategy is marked in bold and the second best is underlined. For Bongard-RWR, we also report the percentage of correctly solved problems in parentheses. For comparison, the average human performance on Bongard-RWR is 39.2 correct answers (65%).

	SYNTHETIC			HOI			OPENWORLD			BONGARD-RWR		
	DI	DE	CO	DI	DE	CO	DI	DE	CO	DI	DE	CO
GPT-4o	17	17	10	35	42	18	40	46	<u>19</u>	5 (8.3%)	<u>8</u> (13.3%)	2 (3.3%)
GPT-4 TURBO	6	15	8	22	45	5	21	57	12	1 (1.7%)	5 (8.3%)	0 (0.0%)
GEMINI 1.5 PRO	7	21	17	23	40	<u>15</u>	13	32	11	3 (5.0%)	7 (11.7%)	1 (1.7%)
CLAUDE 3.5 SONNET	<u>13</u>	<u>19</u>	<u>15</u>	5	<u>44</u>	13	10	<u>53</u>	21	1 (1.7%)	13 (21.7%)	2 (3.3%)
INTERNVL2-8B	0	0	0	12	2	2	11	18	7	0 (0.0%)	0 (0.0%)	0 (0.0%)
LLAVA-1.6 MISTRAL-7B	0	1	0	5	4	1	12	16	1	0 (0.0%)	0 (0.0%)	0 (0.0%)
PHI-3.5-VISION	0	2	0	1	4	2	7	12	5	0 (0.0%)	0 (0.0%)	0 (0.0%)
PIXTRAL 12B	1	4	1	<u>28</u>	27	7	<u>33</u>	34	14	1 (1.7%)	1 (1.7%)	0 (0.0%)

Pixtral stands out among open-access models. Nevertheless, binary classification tasks do not fully reveal whether the solver truly grasped the presented concept or simply relied on surface-level similarities, leaving the space for other challenging evaluation setups.

Generative capabilities in the Direct setting. As presented in Table 1, model performance using the Direct generation strategy is generally weak on synthetic BPs, with the best model, GPT-4o, solving only 17 out of 100 problems. This indicates that the models struggle to identify abstract, synthetic concepts and express them in natural language. The challenge is even more apparent on Bongard-RWR, where the best model, GPT-4o, solves only 5 out of 60 problems. Nevertheless, performance improves on Bongard HOI and Bongard-OpenWorld, with best results of 35/100 and 40/100, resp. Notably, while GPT-4o achieves the highest scores on these two datasets, Pixtral 12B ranks second (28/100 and 33/100, resp.), showing that smaller open-access models can still be competitive in this setting. While the better performance on Bongard HOI and Bongard-OpenWorld may be attributed to a higher ratio of real-world images in the training data, the weak results on Bongard-RWR suggest that the discrepancy is more related to the **specific underlying concepts** than the visual domain as such (see Fig. 14 in Appendix F for the details).

Independent image description. In the next experiment we analyse whether an iterative reasoning approach, in which the model first generates separate captions for each image and then combines them into a final answer, can improve performance. As shown in Table 1, compared to the Direct strategy, the Descriptive one improves the best results across all datasets: from 17 to 21 on synthetic BPs, from 35 to 45 on Bongard HOI, from 40 to 57 on Bongard-OpenWorld, and from 5 to 13 on Bongard-RWR. A clear improvement can be observed individually for each proprietary model. The gain is less pronounced (if at all) for individual open-

access models. We hypothesize that this improvement is due to the Descriptive setting being more aligned with the model’s training, where it primarily learns to caption individual images. However, this strategy doesn’t leverage the additional information present in the joint BP image, and certain context-dependent visual features may be missed in captioning. We believe that with further advancements in reasoning over multi-part compositional images, models in the Direct setting should eventually outperform the Descriptive strategy.

Contrastive reasoning. Correct identification of concepts in BPs requires a joint processing of the images from both sides of the problem. The Contrastive strategy evaluates the model ability to extract underlying differentiating concepts within such image pairs. Across all datasets, the models evaluated under the Contrastive strategy perform worse than with the Descriptive strategy (cf. Table 1). This points to the fundamental difference between human and machine approaches to solving AVR tasks. Humans often rely on direct comparisons between image panels from different categories to highlight differences (Nüssli et al., 2009), whereas the tested methods perform better when making comparisons on text-based image descriptions, potentially disregarding critical visual details missed during image captioning. This discrepancy indicates the need for further modeling improvements to fully leverage the Contrastive strategy.

Iterative reasoning. Next, we tested whether preserving responses from past turns in the dialog context could improve concept identification in both Descriptive and Contrastive settings. As shown in Fig. 5, Descriptive-iterative strategy visibly worsens the results compared to its non-iterative counterpart across all datasets and models, except for negligible improvement of InternVL2-8B on synthetic BPs and several cases of a complete failure (accuracy of 0) of both strategies. In contrast, Contrastive-iterative brings no improvement over Contrastive in only 5 cases, 2 of syn-

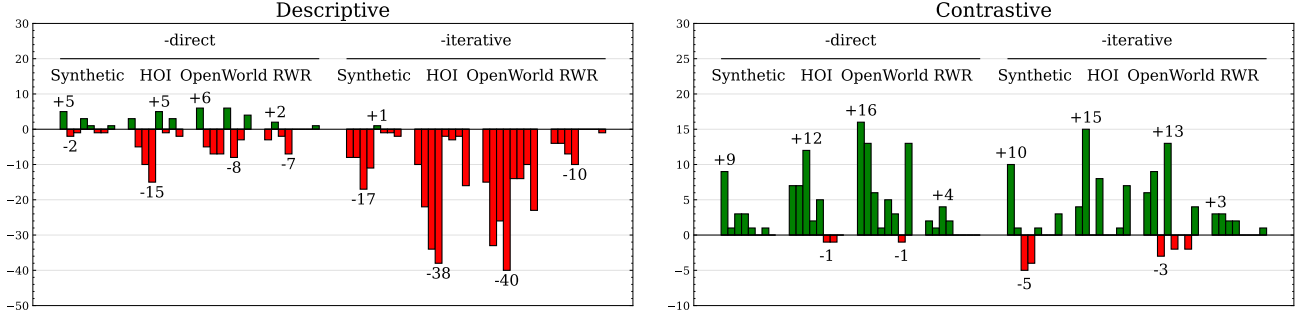


Figure 5: **The impact of -direct and -iterative variants.** Bars in each group correspond to models in the following order: GPT-4o, GPT-4 Turbo, Gemini 1.5 Pro, Claude 3.5 Sonnet, InternVL2-8B, LLaVA-1.6 Mistral-7B, Phi-3.5-Vision, and Pixtral 12B.

thetic BPs, and 3 of Bongard-OpenWorld. Despite these improvements, Contrastive-iterative is generally inferior to Descriptive (see Table 4, Appendix C). Overall, the study indicates that contemporary models struggle to effectively utilize additional information from the context window.

Multimodal answer generation. In the final experiment, we assessed whether incorporating an image of the entire matrix at the answer generation step would improve the performance of the Descriptive and Contrastive strategies. As shown in Fig. 5, Descriptive-direct shows performance improvements over Descriptive in 12 out of 32 (dataset, model) cases. Contrastive-direct improves upon Contrastive in all (dataset, proprietary model) configurations, and additionally improves in certain (dataset, open-access model) settings. However, despite these gains, Contrastive-direct overall performs worse than Descriptive, except for GPT-4o and InternVL2-8B on synthetic BPs, and InternVL2-8B on Bongard HOI (see Table 4, Appendix C). This suggests that contemporary models are to some extent capable of utilizing additional visual inputs to improve reasoning performance, in particular the newest GPT-4o, which displays improvement from incorporating the -direct extension in 7 out of 8 cases. Nevertheless, further work is needed to improve consistency across all models.

Comparison of prompting strategies. Across all models, the Descriptive strategy achieves the highest scores on Bongard-RWR and Bongard-OpenWorld. In Bongard HOI, it ties with Descriptive-direct, while in synthetic BPs, it ranks just behind its -direct extension. As shown in Appendix G, altogether Descriptive strategies solve the same number of synthetic BPs as Contrastive strategies (44; Fig. 16), but lead in Bongard HOI (82 vs. 63; Fig. 18), Bongard-OpenWorld (90 vs. 76; Fig. 20), and Bongard-RWR (20 vs. 11; Fig. 22). This overall advantage of Descriptive over Contrastive strategies indicates that current MLLMs perform better with prompting strategies focused on processing single images. This also highlights the need to

improve multi-image reasoning capabilities of MLLMs for tasks that require reasoning across multiple images. Figs. 16 – 23 in Appendix G further show that altogether the considered approaches solved 54, 89, 93, and 23 problems from synthetic BPs, Bongard HOI, Bongard-OpenWorld, and Bongard-RWR, resp. This raises the question of whether an ensemble combining all proposed strategies could further enhance model reasoning performance. We leave the exploration of this emerging direction for future work.

Proprietary vs. open-access models. Proprietary models generally outperform open-access ones, leading in 35 out of 40 (dataset, strategy) pairs (see Appendix C, Table 4). The black-box nature of proprietary models makes it challenging to attribute their advantage to specific aspects, whether it be the number of parameters, the size and composition of training data, or the pre- and post-processing methods. However, the recently released Pixtral 12B model performs competitively in multiple settings, occasionally surpassing proprietary models. This highlights the viability of developing competitive MLLMs without sacrificing accessibility. At the same time, a clear performance drop of Pixtral 12B on synthetic BPs and Bongard-RWR suggests its intrinsic weakness in reasoning about *abstract* concepts, whether reflected in synthetic or real-world manner, similarly to other open models.

Dataset variants. We evaluated the four open-access models on the three binary classification setups presented in Section 3.3 using the additional Bongard-RWR variants. In the Ground-Truth and Incorrect Label settings, all models performed similarly in the original Bongard-RWR and its three new variants. However, in the Images to Sides setting, model performance differed. On RWR-G, most models solved more problems compared to Bongard-RWR, e.g., Phi-3.5-Vision (+15), suggesting that removing color helped models focus on structural concepts. The difference was also present for RWR-S, e.g., for InternVL2-8B (+13), Pixtral 12B (+9), and Phi-3.5-Vision (+7), likely due to

cropping, which increased the signal-to-noise ratio by eliminating empty space between panels. RWR-SG followed a similar pattern. Overall, these additional dataset variants not only make the task more approachable but also help measure robustness to matrix layout and sensitivity to image color. Extended results are provided in Appendix C.

Concept selection. Similarly to the evaluation setup proposed in (Wüst et al., 2024), we conducted a multi-class classification experiment in which the MLLM selects the correct concept from a list of $k \in \{2, 4, 8, 16\}$ choices. Negative candidates were sourced from subsequent matrices, ensuring they were semantically different from the correct option. Across all four datasets, models achieved the highest accuracy on Bongard HOI and Bongard-OpenWorld (exceeding 90% in most cases), while synthetic BPs and Bongard-RWR posed a greater challenge (e.g., for InternVL2-8B, 16% and 25% resp., for $k = 16$). These results suggest that real-world concepts, such as “*Swing tennis racket vs. Not swing tennis racket*”, are easier for the models to recognize, likely because they can rely on surface-level object or action recognition to exclude incorrect options. In contrast, synthetic BPs and Bongard-RWR demand recognizing abstract concepts, such as “*Convex vs. Nonconvex*”, which require a deeper understanding of the images. The gap between Bongard-RWR and synthetic BPs further highlights the additional challenge posed by representing abstract concepts in real-world images. The detailed results are provided in Appendix C.

Model scaling. We investigated the effect of model scaling on abstract reasoning capabilities by evaluating a range of open-access and proprietary models of varying sizes across all four datasets using both the Direct and Descriptive strategies. Our results reveal that while larger models generally perform better, especially among open-access models, smaller variants occasionally outperform their larger counterparts. Nevertheless, even smaller proprietary models (e.g., GPT-4o mini) consistently outperformed the largest open-access alternatives, suggesting that scaling alone may be insufficient to achieve strong abstract reasoning capabilities. This highlights the need for explicit, dedicated efforts to improve abstract reasoning skills of MLLMs in future work. Appendix D provides an in-depth analysis of these findings.

Comparison with state-of-the-art. A direct comparison with the results from (Wu et al., 2024) is challenging due to the different ranges of test problems used in each study. With this caveat, we concentrate on key high-level observations from both works. Wu et al. (2024) primarily focus on a binary classification setting corresponding to the Images to Sides setup in our work. On Bongard-OpenWorld, our best performing models, Gemini 1.5 Pro and Claude 3.5 Sonnet, achieved 96% accuracy, while their top method, SNAIL—a meta-learning approach leveraging pre-trained OpenCLIP

image representations—achieved 64%. This suggests that MLLMs, which uniformly process images and text, outperform decoupled two-stage approaches, which handle image captioning and text-based reasoning with different models. They also briefly consider a natural language generation task, where models describe concepts presented in the BP instance (Wu et al., 2024, Appendix E). They again use a two-step approach comprising fine-tuned BLIP-2 for image captioning and ChatGPT for concept generation. In contrast, we employ a single MLLM for both tasks. For evaluating free-form concept generation, they use automated text-based metrics, which provide a general measure of text similarity. We, however, employ a voting MLLM ensemble, offering a more direct assessment of solution correctness.

Human results. We conducted a study on Bongard-RWR with 30 participants, as detailed in Appendix B. Participants were asked to describe BP concepts in an open-ended manner. Their responses were evaluated by expert annotators, who compared them to ground-truth labels and classified as correct or incorrect. Humans solved from 23 to 59 problems, with average of 39.2, achieving 65% accuracy. Notably, the lowest number of problems solved by a human participant (23) exceeded the number of problems solved by all models in total (22, see Fig. 23), which signifies a clear performance gap.

6. Conclusions

This paper investigates the reasoning capabilities of proprietary and open-access MLLMs using BPs as a case study. Despite rapid progress, MLLMs still exhibit significant reasoning limitations. Across all proposed answer generation strategies, the best-performing model solved only 22 out of 100 synthetic BPs. On the other hand, model performance improved moderately with real-world concepts, as shown by the results on Bongard HOI and Bongard-OpenWorld. To delve deeper into the performance discrepancies between synthetic and real-world domains, we introduced Bongard-RWR, a new BP dataset designed to represent concepts from synthetic BPs via real-world images. Focused experiments with this dataset suggest that the models’ weak performance on synthetic BPs is not domain-specific but rather indicative of broader limitations in their reasoning abilities. Specifically, MLLMs struggle with recognizing abstract concepts, fail to benefit from a human-like multi-image reasoning approach, demonstrate limitations in utilizing context window effectively, and require further work to consistently integrate text and vision modalities at the answer generation step. On a positive note, experiments conducted in three binary classification settings show that some models achieve encouraging results, suggesting that current limitations may be overcome with future advancements.